



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO M S P AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

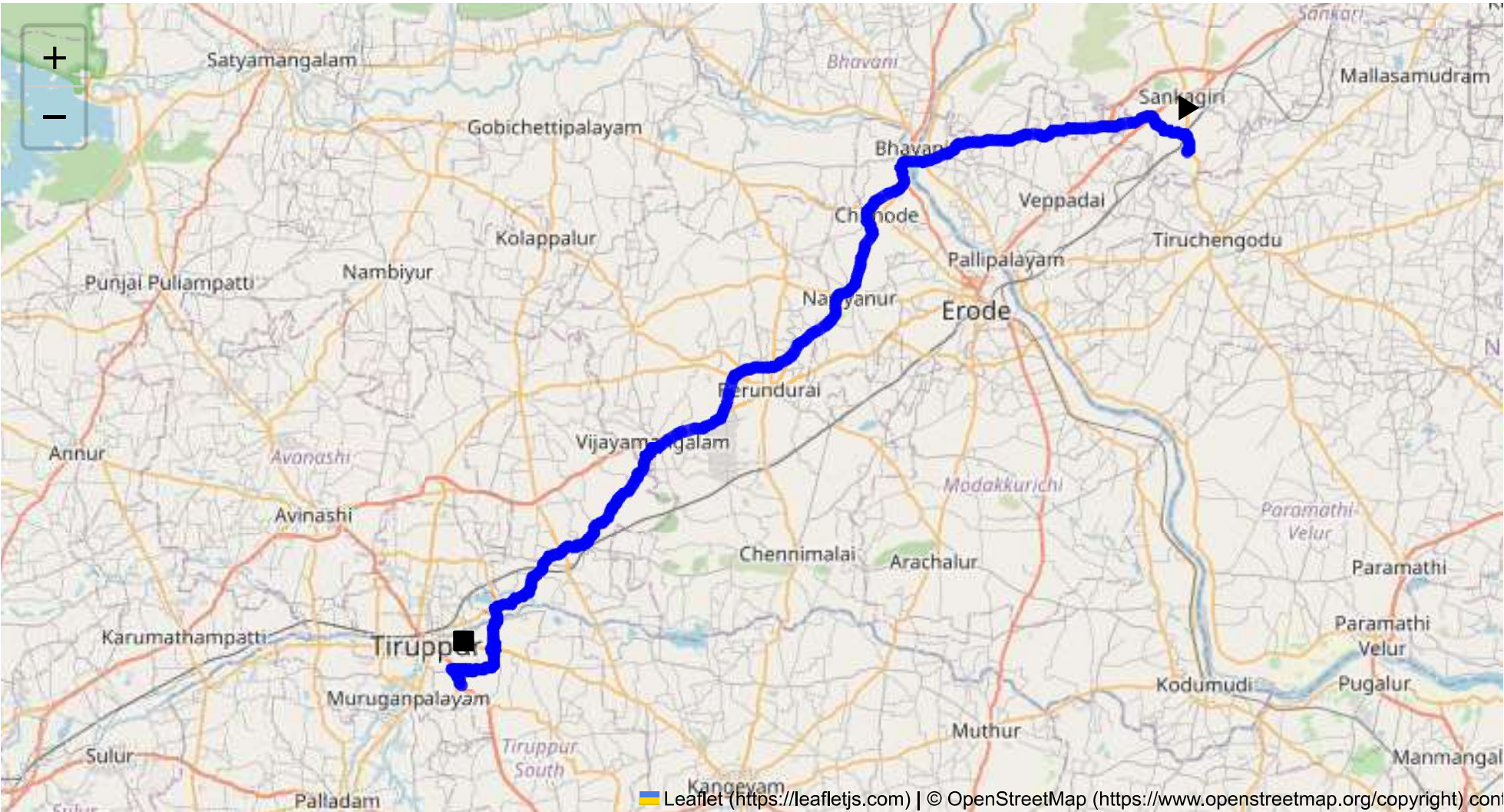
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 85.27 km**  
**Estimated Duration: 1.9 hours**  
**Adjusted Duration (Heavy Vehicle): 2.3 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.075189, 77.369332)**



Welcome to the Journey Risk Management Study

Analyzing the route from CVQF+23W, Sangagiri to 39G9+6J4, K Chettipalayam, involves several aspects to ensure safe transportation, especially when dealing with heavy vehicles carrying hazardous materials. Here is a comprehensive analysis of the route:

1. Overview of the Route Map:

The route begins at Sangagiri, heading southeast through rural and semi-urban areas before reaching



Tiruppur. The journey covers state highways and possibly some national highways, traversing through villages, agricultural lands, and small towns. The road network is diverse, connecting various small communities, with Sangagiri lying on NH-47 leading towards Tiruppur in Tamil Nadu.

**2. Typical Weather Conditions and Potential Weather-Related Hazards:**

Tamil Nadu experiences a tropical climate with hot summers, mild monsoons, and occasional cyclones, especially from October to December. The potential weather hazards include heavy rains during the monsoon, which can lead to flooding and poor visibility. Fog and mist are rare but possible during early mornings in winter months.

**3. Analysis of Traffic Patterns:**

Traffic is generally moderate but can become heavy during peak hours, typically in the morning (8 AM - 10 AM) and evening (5 PM - 8 PM). Congestion is more likely around towns such as Erode and near textile hubs approaching Tiruppur. Infrastructure improvements or roadworks could also temporarily increase congestion.

**4. Assessment of Road Quality and Infrastructure:**

The road conditions vary; state highways may have uneven surfaces and fewer lanes compared to national highways. Potholes and occasional road maintenance can cause disruptions. However, recent infrastructure development might have improved certain segments.

**5. Suggestions for Alternative Routes for Emergencies:**

In case of roadblocks or emergencies on the main route, consider taking alternative state roads that run parallel or adjacent to the primary thoroughfare. Local roads connecting between NH-47 and SH levels should be pre-identified as part of contingency planning.

**6. Summary of Local Regulations Affecting Hazardous Material Transport:**

Transport of hazardous materials is governed by strict regulations, including permitted transit times, mandatory rest stops, and routes approved by local transport authorities. Compliance with safety standards for securing HAZMAT cargo is essential.

**7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:**

There have been occurrences of accidents due to overloaded trucks and non-compliance with speed limits. Check local transport authority databases for specifics regarding any major incidents and adapt safety measures accordingly.

**8. Environmental Considerations and Sensitive Areas:**

The route includes agricultural zones and possibly protected lands. Environmental regulations pertain to emissions and noise levels, requiring adherence to pollution control norms. Avoiding spills and managing waste responsibly in these areas is crucial.

**9. Analysis of Communication Coverage:**

Generally, this region is well-covered by mobile networks; however, there might be occasional dead zones in rural stretches and near forested areas. Drivers should ensure communication devices are fully charged and consider satellite communications as a backup.

**10. Estimated Emergency Response Times for Different Route Segments:**

Emergency services, like ambulances or fire brigades, are more accessible closer to urban centers like

Erode and Tiruppur, with longer response times expected in isolated rural areas. It’s advisable to coordinate with local emergency services for quicker assistance.

11. Overall Summary of Risk Assessment:

The route from Sangagiri to K Chettipalayam poses moderate risk, primarily from potential weather disruptions, variable traffic conditions, and road quality. Adhering to safe driving practices, maintaining compliance with transport regulations, and preparing for emergencies are essential for minimizing risks. Ensuring communication availability and understanding local regulations will further contribute to a safer transit for hazardous materials.

This analysis equips drivers with vital information to mitigate potential risks and navigate safely along this route. Being proactive in preparation and aware of the topographical, regulatory, and environmental factors enhances transportation safety and efficiency.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44024, 77.87528 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44898, 77.87410 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.44965, 77.77565 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.44953, 77.77344 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.43165, 77.67734 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.23724, 77.50686 | 30 KM/Hr    |
| 11 | Turn       | High       | 11.23704, 77.50676 | 15 KM/Hr    |
| 12 | Blind Spot | Blind Spot | 11.23686, 77.50216 | 10 KM/Hr    |
| 13 | Turn       | Medium     | 11.23649, 77.50219 | 30 KM/Hr    |
| 14 | Turn       | High       | 11.23396, 77.50105 | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.23287, 77.49934 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.18407, 77.46330 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.18273, 77.46224 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.17851, 77.46169 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.17521, 77.45978 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 20 | Turn       | Medium     | 11.17368, 77.45730 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.17111, 77.45547 | 30 KM/Hr    |
| 22 | Turn       | High       | 11.16959, 77.44396 | 15 KM/Hr    |
| 23 | Turn       | Medium     | 11.16381, 77.43700 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.16280, 77.43139 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.15217, 77.42418 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.15134, 77.42300 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.15023, 77.42247 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.14925, 77.42065 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.13848, 77.41611 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.13578, 77.41299 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.12989, 77.40302 | 30 KM/Hr    |
| 32 | Turn       | High       | 11.12775, 77.39347 | 15 KM/Hr    |
| 33 | Turn       | High       | 11.11846, 77.39460 | 15 KM/Hr    |
| 34 | Turn       | High       | 11.11833, 77.39390 | 15 KM/Hr    |
| 35 | Turn       | High       | 11.10276, 77.39199 | 15 KM/Hr    |
| 36 | Turn       | Medium     | 11.10256, 77.39273 | 30 KM/Hr    |
| 37 | Blind Spot | Blind Spot | 11.09882, 77.39312 | 10 KM/Hr    |
| 38 | Turn       | High       | 11.09898, 77.39028 | 15 KM/Hr    |
| 39 | Turn       | Medium     | 11.09456, 77.39233 | 30 KM/Hr    |
| 40 | Turn       | High       | 11.08759, 77.39166 | 15 KM/Hr    |
| 41 | Turn       | Medium     | 11.08531, 77.38413 | 30 KM/Hr    |
| 42 | Turn       | Medium     | 11.08597, 77.38095 | 30 KM/Hr    |
| 43 | Turn       | High       | 11.08536, 77.37919 | 15 KM/Hr    |
| 44 | Turn       | Medium     | 11.08481, 77.37902 | 30 KM/Hr    |
| 45 | Turn       | Medium     | 11.08448, 77.37842 | 30 KM/Hr    |
| 46 | Turn       | Medium     | 11.08510, 77.37541 | 30 KM/Hr    |
| 47 | Turn       | Medium     | 11.08463, 77.37378 | 30 KM/Hr    |
| 48 | Turn       | High       | 11.08497, 77.37234 | 15 KM/Hr    |
| 49 | Turn       | High       | 11.08544, 77.37228 | 15 KM/Hr    |
| 50 | Turn       | High       | 11.08604, 77.36377 | 15 KM/Hr    |
| 51 | Turn       | High       | 11.08583, 77.36360 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                           | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------------------|---------------------------|-------------|------------|
| 3  | hospital | Dhanvanthri Hospital                           | 11.4496712,<br>77.7593772 | 30 km/h     | Medium     |
| 4  | hospital | J.K.K. Trust Hospital                          | 11.4445841,<br>77.7307962 | 30 km/h     | Medium     |
| 6  | hospital | Shri Sathyanarayana Hospital                   | 11.4291297,<br>77.6913408 | 30 km/h     | Medium     |
| 7  | hospital | Thanish Siddha Hospital                        | 11.430003,<br>77.674964   | 30 km/h     | Medium     |
| 8  | clinic   | Harshitha Clinic                               | 11.4313207,<br>77.674718  | 30 km/h     | Medium     |
| 9  | hospital | Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam | 11.432369,<br>77.674894   | 30 km/h     | Medium     |
| 10 | clinic   | G.K Clinic                                     | 11.4297244,<br>77.6749715 | 30 km/h     | Medium     |
| 11 | clinic   | Erode Cancer Centre                            | 11.3732, 77.649152        | 30 km/h     | Medium     |
| 13 | hospital | Irt Hospital                                   | 11.2803603,<br>77.5644118 | 30 km/h     | Medium     |
| 15 | hospital | Gen Siddha Hospital                            | 11.2375033,<br>77.5059797 | 30 km/h     | Medium     |
| 16 | hospital | Vijayamangalm Government Hospital              | 11.2382827,<br>77.501687  | 30 km/h     | Medium     |
| 18 | hospital | Aravind Eye Care Hospital                      | 11.1729469,<br>77.4574467 | 30 km/h     | Medium     |
| 19 | clinic   | Sri Maha Annai Clinic                          | 11.1707162,<br>77.4536498 | 30 km/h     | Medium     |
| 20 | hospital | Ram Doctor Hospital                            | 11.1709391,<br>77.4532808 | 30 km/h     | Medium     |
| 21 | clinic   | Kumaran Clinic                                 | 11.1697553,<br>77.451543  | 30 km/h     | Medium     |
| 22 | hospital | Sri Aravindhar Hospital                        | 11.1697756,<br>77.4448276 | 30 km/h     | Medium     |
| 23 | clinic   | Dr. Raja Clinic                                | 11.1278464,<br>77.3927768 | 30 km/h     | Medium     |
| 24 | clinic   | D K Clinic General & Ortho                     | 11.1117864,<br>77.3906835 | 30 km/h     | Medium     |



|    | type     | name                          | coordinates              | speed_limit | risk_level |
|----|----------|-------------------------------|--------------------------|-------------|------------|
| 25 | hospital | Tiruppur, Government Hospital | 11.083827,<br>77.3634797 | 30 km/h     | Medium     |

Crowded Spots

|    | type        | name                                      | coordinates               | speed_limit | risk_level |
|----|-------------|-------------------------------------------|---------------------------|-------------|------------|
| 0  | school      | KRP Matric. Hr. Sec School                | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 1  | college     | Vivekanandha Engineering College          | 11.4589312,<br>77.7899284 | 30 km/h     | Medium     |
| 2  | marketplace | Monday market                             | 11.452863,<br>77.775989   | 30 km/h     | Medium     |
| 5  | school      | SSM Matriculation Higher Secondary School | 11.4321653,<br>77.6880046 | 30 km/h     | Medium     |
| 12 | college     | Government polytechnic college            | 11.2907053,<br>77.5698726 | 30 km/h     | Medium     |
| 14 | school      | Bharathi Matriculation School             | 11.2494613,<br>77.5307028 | 30 km/h     | Medium     |
| 17 | college     | Sasurie Engineering College               | 11.2171482,<br>77.4898552 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Turn  
Risk Level: High



Speed Limit: 15 KM/Hr  
Coordinates: 11.44024, 77.87528



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.44898, 77.87410



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.45352, 77.85698





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45898, 77.85135





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44965, 77.77565



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44953, 77.77344





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.43165, 77.67734



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.23724, 77.50686





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.23704, 77.50676



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23686, 77.50216



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23649, 77.50219





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.23396, 77.50105



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.23287, 77.49934





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.18407, 77.46330



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.18273, 77.46224



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.17851, 77.46169





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.17521, 77.45978



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.17368, 77.45730





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.17111, 77.45547



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.16959, 77.44396



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16381, 77.43700





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16280, 77.43139



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.15217, 77.42418





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.15134, 77.42300



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.15023, 77.42247



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.14925, 77.42065





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.13848, 77.41611



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.13578, 77.41299





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.12989, 77.40302



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.12775, 77.39347



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.11846, 77.39460





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.11833, 77.39390



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.10276, 77.39199





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.10256, 77.39273



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.09882, 77.39312



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.09898, 77.39028





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.09456, 77.39233



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.08759, 77.39166





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08531, 77.38413



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08597, 77.38095



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.08536, 77.37919





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08481, 77.37902



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08448, 77.37842





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08510, 77.37541



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08463, 77.37378



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.08497, 77.37234





Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.08544, 77.37228



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.08604, 77.36377





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.08583, 77.36360

